

Sukhrob Ilyosbekov

Machine Learning Engineer · Computer Vision & LLM Systems

📍 Portland, ME 📞 (857) 867-1942 ✉️ silyosbekov@gmail.com

🌐 suxrobgm.net/research 🎓 [Google Scholar](https://scholar.google.com/citations?user=suxrobgm) 🔗 linkedin.com/in/suxrobgm 🐙 github.com/suxrobgm

SUMMARY

I work on every stage of an ML system, from data prep and training to serving and monitoring. My research is in computer vision, mostly medical imaging and representation learning, with three arXiv preprints. Nine years as a software engineer made the production side familiar. Background in applied math.

TECHNICAL SKILLS

Deep Learning & CV: PyTorch, TensorFlow, CNNs, Vision Transformers, DINOv3, EfficientNet, YOLO, contrastive learning, GradCAM++, OpenCV

LLM Systems: Claude/GPT/DeepSeek APIs, tool-use agents, MCP, RAG, vision-language models, Hugging Face

Math & Modeling: Probability and statistics, linear algebra, optimization, time-series forecasting, feature engineering, A/B testing, error analysis

Languages: Python, C#, C++, TypeScript, Kotlin

Data & Serving: NumPy, pandas, scikit-learn, FastAPI, PostgreSQL, SQL Server, MongoDB, Redis, OHIF/DICOM

Infrastructure: CUDA, mixed-precision and multi-GPU training, model serving and drift monitoring, Docker, Kubernetes, GitHub Actions, AWS, Azure

Software: ASP.NET Core, SignalR, Blazor, Node.js, Bun, NestJS, React, Next.js, Angular

EDUCATION

Northeastern University, M.S. in Computer Science, GPA 4.0, Boston, MA *Jan 2024 – May 2026*

Suffolk University, B.S. in Computer Science, Cum Laude, Boston, MA *Sep 2021 – May 2023*

Honors: National Physics Olympiad, 1st Place (Uzbekistan) · Associate Member, Sigma Xi · Coding Competition Winner, TUIT

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | [EmTech Care Labs](#) | Portland, ME *Jan 2025 – Present*

- Built a model that reads Alzheimer's residents' daily activity and caregiver logs and flags who is at risk of decline, so staff can step in before a crisis.
- Wrote an NLP pipeline that turns free-text clinical notes into structured records of symptoms, medications, and care events. It replaced manual chart review.
- Both models run in production on de-identified data under HIPAA, with nightly retraining, drift alerts, encrypted PHI, and an audit trail on every prediction.
- Also lead engineering for the care platform itself, a TypeScript monorepo on AWS that passed an outside HIPAA readiness review with no critical findings.

Machine Learning Engineer | [AllFactors](#) | Santa Clara, CA *Oct 2021 – Dec 2024*

- Built the property-valuation and market-forecast models that replaced the company's rules-based system.
- Wrote the real-time feature pipeline behind them on SignalR and SQL Server change tracking. New transactions reached the models within seconds and fed a nightly retrain.
- Users saw predictions in Blazor WebAssembly dashboards; automated tests covered the whole path from incoming data to prediction.

Earlier: Software Engineer at Smart Meal Service, Russia (2020 – 2021), writing C# software for food kiosks and a robotic cashier. Game Developer at Pentalight Technology, Malaysia (2020 – 2021), on networking and hand tracking for a Unity VR simulation. Freelance Python NLP and .NET work (2019 – 2020). Game Developer at EC Dev Team, Uzbekistan (2016 – 2019).

SELECTED AI/ML PROJECTS

MorphoCLIP

[GitHub](#) | [Paper](#)

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Matches Cell Painting microscopy of cells treated with drugs or genetic perturbations to plain-language descriptions of the treatment, using contrastive image-text training.
- The DINOv3 and BioClinical ModernBERT backbones stay frozen and only the projection heads train, so it fits on one consumer GPU.
- Evaluated on CPJUMP1 (51 plates, 3M+ images). As first author I led design, training, and evaluation.

Localize, Don't Beautify

<https://arxiv.org/abs/2608.02841>

Image-editing APIs, inpainting, ArcFace, prompt engineering

- Commercial image editors asked for one surgical change tend to retouch the whole face. I tested prompt steering, masked compositing, and inpainting as client-side fixes across six editors on 196 preview edits.
- Scored identity preservation with ArcFace and checked whether each edit stayed in its region. Plain masked compositing kept edits in place best. Sole author.

MelanomaNet

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- Nine-class skin-lesion classifier on ISIC 2019 (25K images). EfficientNet V2 with focal loss to handle the class imbalance; 0.86 weighted F1.
- Scores the model's GradCAM++ attention against the ABCDE criteria dermatologists use, each measured from the lesion mask. Sole author.

Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- Pulls X-ray, CT, and MRI studies from hospital imaging systems over DICOM and runs PyTorch detectors on them, such as pneumonia on chest X-rays and hemorrhage on head CTs.
- Results show up as overlays in OHIF. Studies are de-identified on the fly, and access is role-based and logged.

LogisticsX

<https://logisticsx.app> | [GitHub](#)

Multi-provider LLMs, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin Multiplatform, PostgreSQL

- A multi-provider LLM agent with custom tools does the dispatcher's job: it assigns freight to trucks, checks federal hours-of-service rules, and plans multi-stop routes.
- I built the rest of the product as well: a multi-tenant .NET backend, Angular portals, a Kotlin Multiplatform driver app, and load-board and telematics integrations.

Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

YOLO, Moondream2 VLM, llama.cpp, FastAPI, Angular

- Photograph a shelf and get a book list back. YOLO segments each spine, Moondream2 reads the title and author, and an Angular UI handles corrections and export. Won an Outstanding Project Award at Northeastern.

OGStack

<https://ogstack.dev> | [GitHub](#)

Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe

- Uses Claude, GPT, and DeepSeek to read a site's branding and generate its link-preview image. Served 10K+ images from a global CDN during beta.